DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II   Advanced Calculus (2AC)

Tutorial Sheet 9: **20th and 21st April 2023**

1. Say whether the following statement is true or false. Explain clearly your answer.

   Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function with continuous partial derivatives up to order two. If a critical point (a, b) of the function f is such that

   $$f_{xx}(a,b) > 0 \quad \text{and} \quad f_{xx}(a,b) + f_{yy}(a,b) = 0\,,$$

   then the point (a, b) is a point of local minimum for the function f.

2. Given the function $f(x, y) := xy + x - y$.

   (i) Determine the nature of each critical point of f.

   (ii) Find the points of minimum and maximum of f over the circle $x^2 + y^2 = 25$.

3. Find the points of maximum and minimum of the function $f(x, y) := -x + y^2$ over the ellipse $E := \{(x, y) \in \mathbb{R}^2\colon\ x^2 + 4y^2 = 1\}$.

4. Consider the function $f(x, y, z) = x + y + z$. Find the maximum and minimum values of f on the sphere $x^2 + y^2 + z^2 = 1$.

5. A box has to be constructed out of 120 cm^2 of material. Find the dimensions x, y and z of the box that maximize the volume.
   (Hint: For the dimensions x, y and z, find the volume, the total area and hence, set-up the corresponding optimization problem and solve it.)

6. **(After Max/min-IV to be covered in the lecture on Friday)**
   Given the function $f(x, y) := xy + x - y$. Let $D := \{(x, y) \in \mathbb{R}^2\colon\ x^2 + y^2 \leq 25 \text{ and } x \geq 0\}$. Find the points of maximum and minimum of f on the domain D.